

NAGESWARA RAO VUTLA

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SUMMARY

Machine Learning Engineer with expertise in **NLP, Generative AI, Computer Vision, and end-to-end ML pipelines**. Skilled in **Python, TensorFlow, PyTorch**, and cloud platforms like **AWS SageMaker**. Experienced in building **AI-driven solutions** that optimize business processes and enhance decision-making. Passionate about **LLMs, AI Agents, and deploying scalable ML models**.

EDUCATION

Bapatla Engineering College, Bapatla

Bachelor of Technology in Mechanical Engineering | April 2023 | **GPA: 8.23/10**

PROFESSIONAL EXPERIENCE

Machine Learning Intern

Quixy, Vividminds Technology | Jul 2024 - Present

- Developed AI solutions for **NLP and predictive modeling**, improving **data analysis efficiency by 20%**.
- Built **end-to-end ML pipelines** (Python, TensorFlow, PyTorch) from **data ingestion to model deployment**.
- Enhanced business intelligence with **RAG, AI Agents, and NLP-powered document processing**.
- Optimized SQL-based **data retrieval using ChromaDB and vector search**, reducing query execution time.

PROJECTS

Governance Decision System

- Tech Stack:** GPT-4, ChromaDB, SQL, Python
- Designed an **AI-powered dashboard** for **data-driven governance**, integrating **vector search & predictive modeling**.
- Improved **decision-making efficiency by 25%** through automated SQL-based analysis.

Multi-Doc QA System

- Tech Stack:** GPT, LLaMA3, LangChain, Tesseract OCR
- Built an **AI-based document QA system** for querying **PDFs, Word, Excel, and web pages**.
- Achieved **90% accuracy in document parsing** by integrating OCR and multi-modal retrieval.

AI-driven HRMS System

- Tech Stack:** CrewAI, LangChain, Python
- Developed an **AI-powered hiring automation tool** with **resume evaluation, AI interviews, and identity verification**.
- Reduced hiring time by **30%** through automated screening and Aadhaar-based verification.

Number Plate Detection and OCR Extraction

- Tech Stack:** YOLOv8, OpenCV, Tesseract OCR
- Built a **real-time vehicle number plate detection system** with **OCR-based text extraction**.
- Achieved **95% accuracy**, enhancing **traffic monitoring and law enforcement applications**.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Data Manipulation & Analysis: NumPy, Pandas

Data Visualization: Matplotlib, Seaborn, Plotly

Machine Learning & AI: Core Expertise in **NLP, Deep Learning, Generative AI, Computer Vision, Model Fine-Tuning, Prompt Engineering**

LLM & AI Agents: RAG (Retrieval-Augmented Generation), Agentic RAG, AI Agents, Multi-Agent Systems

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, LangChain, LlamaIndex, CrewAI, LangGraph, AutoGen

Tools & Technologies: Docker, Git, Tesseract OCR, AWS SageMaker, Vector Databases (ChromaDB, FAISS)